

IAL Physics Unit 2 — Electrical Energy, Power & Heating | Recall Pack

IAL Physics Unit 2 — Electrical Energy, Power & Heating

The whole topic in one mental model. Electrical power is just *energy delivered per second* — and there are three faces of the same idea: $P = VI$ (the master equation, always true), $P = I^2R$ (the heating-effect form, for what's wasted as heat), and $P = V^2/R$ (the voltage-driven form). They are not three separate laws — substitute Ohm's law $V = IR$ into $P = VI$ and the other two pop out. The microscopic story is electrons colliding with the lattice ions, making them vibrate (= heat). Multiply any power formula by time and you get energy ($W = Pt = VIt = I^2Rt$). The whole topic is: pick the right power formula for the quantities you're given, never take the roundabout route to energy, get fuses right ($I = P/V$ then round up), and quote efficiency as a percentage.

1. Jargon glossary

Term	Student-friendly meaning	Exact physics meaning	Symbols/units	What it is NOT	Lesson example	Phuc/common trap
Power P	How fast energy is being transferred — the energy delivered every second. A 2 kW kettle pours 2000 J of energy into the water each second.	The rate of energy transfer (rate of doing work): $P = \frac{\Delta W}{\Delta t}$. For an electrical device $P = VI$.	P ; watts $W = J\ s^{-1}$	NOT energy itself (energy is $P \times t$); NOT measured in joules; NOT the same as current or voltage.	Alex derives $P = IV$ on the board from $P = W/t$, $I = Q/t$, $V = W/Q$ (Loom a28d777a, 06:30).	Common trap: power (W) and energy (J) are different quantities — a “2000 W” rating is a <i>rate</i> , you still need the time to get joules.

Term	Student-friendly meaning	Exact physics meaning	Symbols/units	What it is NOT	Lesson example	Phuc/common trap
Electrical energy / work done W	The total energy a charge delivers as it's pushed through a p.d. — what you actually pay the electricity company for.	Energy transferred when charge Q moves through p.d. V : $W = QV = VIt = Pt$. The cumulative version of power.	W (or E); joules J	NOT power (that's the <i>rate</i>); the italic W (work, J) is NOT the upright W (watts) — same letter, different meaning.	Kettle: 540 kJ delivered while 2250 C passes $\Rightarrow V = 540 \times 10^3 / 2250 = 240$ V (textbook Ch8.5).	Phuc trap: reach for $W = Pt$ or $W = VIt$ directly — the roundabout $Q = It$ then $E = QV$ wastes two steps and doubles the chance of a unit slip (Alex wrote $E = Pt, E = IVt, E = I^2Rt$ to push the one-step method, a28d777a 10:29).
The master power equation $P = VI$	The one power formula that is <i>always</i> true for any electrical component — current times the voltage across it.	$P = VI$: the general electrical power, derived by dividing $W = QV$ by t (since $W/t = P$ and $Q/t = I$). Works for any component, ohmic or not.	P W; V V; I A	NOT limited to resistors; does NOT require Ohm's law; the other two power forms are <i>derived from</i> this one.	Battery power example: $P_{\text{battery}} = V_{\text{battery}} \times I_{\text{battery}}$, use the battery's own voltage (a28d777a 55:43).	Phuc trap: for battery power use $V_{\text{battery}} \times I_{\text{battery}}$ — Phuc tried $P = 0.83 \times 3.6$ using a resistor's value instead of the battery's (a28d777a 55:43).
Heating-effect formula $P = I^2R$	The power <i>wasted as heat</i> in a resistor — what makes a wire get hot. Use it when you know the current and the resistance.	Substitute $V = IR$ into $P = VI$: $P = I(IR) = I^2R$. At constant temperature this is exactly the rate of heat dissipation in a resistor.	P W; I A; R Ω	NOT the general formula (that's VI); NOT for use when you only have V and R ; $P \propto I^2$, not $\propto I$.	Series resistor: $P = I^2R = (0.75)^2 \times 2.0 = 1.125 \approx 1.1$ W (Worked Ex 1).	Common trap: because $P \propto I^2$, doubling the current quadruples the power — not doubles it.
Voltage-driven formula $P = V^2/R$	The power form to use when you know the voltage across a component and its resistance — no need to find the current first.	Substitute $I = V/R$ into $P = VI$: $P = (V/R)V = V^2/R$.	P W; V V; R Ω	NOT $P = (V/R)^2$; NOT for when current is the known quantity; $P \propto V^2$ at fixed R .	Parallel section at 1.5 V: $P_{6\Omega} = 1.5^2/6 = 0.375$ W, $P_{3\Omega} = 1.5^2/3 = 0.75$ W (Worked Ex 2).	Common trap: at fixed R , doubling the voltage quadruples the power ($P \propto V^2$); at fixed V , doubling R <i>halves</i> the power ($P \propto 1/R$).

Term	Student-friendly meaning	Exact physics meaning	Symbols/units	What it is NOT	Lesson example	Phuc/common trap
Heating effect of a current	Why a current makes a wire hot: drifting electrons keep banging into the metal's fixed ions, shaking them harder — and atoms shaking harder <i>is</i> heat.	Electrons collide with the positive lattice ions, transferring kinetic energy; the ions vibrate with greater amplitude (= increased internal/thermal energy). Rate of heat transfer = $I^2 R$.	rate = $I^2 R$ (W)	NOT the electrons being “stored” in the resistor — charge in = charge out (Kirchhoff 1); NOT the electrons themselves heating up.	Alex: “When they bash into them, they transfer some of their kinetic energy. The ions then vibrate more, and that’s what heat is.” (a28d777a).	Phuc trap: a resistor does NOT store electrons — they’re slowed, not accumulated; charge entering = charge leaving (Trap 7).
Dissipated (energy)	Energy “scattered away” as heat into the resistor’s atoms — lost to the surroundings, not recoverable.	Electrical energy irreversibly converted to thermal (internal) energy of the resistor’s lattice; the resistor’s internal energy rises.	(described as $I^2 R t$ joules)	NOT energy stored; NOT recoverable as electrical energy; NOT “destroyed” — it’s transferred to thermal form.	Heating-effect rate $I^2 R$ is the power dissipated as heat (Core Theory 3.1).	Common trap: “dissipated” means <i>transferred to thermal energy of the surroundings</i> , not “used up / destroyed” — energy is conserved.
Efficiency η	What fraction of the energy you put in actually does the useful job, as a percentage — the rest leaks away (usually as heat).	$\eta = \frac{P_{\text{useful}}}{P_{\text{total}}} \times 100\%$ (equivalently $\eta = W_{\text{useful}}/W_{\text{total}} \times 100\%$). For an electrical device $P_{\text{total}} = VI$.	η ; dimensionless, quoted as a %	NOT greater than 100%; NOT a power or an energy; the “wasted” part is usually heat, not “lost” energy.	Motor / appliance: in 500 J, useful 350 J in 10 s $\Rightarrow \eta = 35/50 \times 100 = 70\%$ (Self-test Q7).	Common trap: always express as a percentage and label which power is “useful” and which is “total”; never quote efficiency above 100%.
Fuse	A deliberately thin wire that melts and breaks the circuit if too much current flows — a one-shot safety cut-out protecting the appliance from fire.	A thin wire with a rated current; if current exceeds the rating it melts (its own $I^2 R$ heating), breaking the circuit. Standard UK ratings: 3 A, 5 A, 13 A.	rated current in A	NOT a resistor you keep in circuit normally; NOT reusable after it blows; does NOT “limit” current smoothly — it’s all-or-nothing.	Kettle 240 V, 3.0 kW $\Rightarrow I = P/V = 12.5 \text{ A} < 13 \text{ A}$, so the 13 A fuse is safe (Worked Ex 3, fuse example).	Phuc trap: fuse current uses $I = P/V$ (Marta first slipped to $I = V/P$, e14ebf09); then choose the next standard rating above the operating current — never below.

Term	Student-friendly meaning	Exact physics meaning	Symbols/units	What it is NOT	Lesson example	Phuc/common trap
Current rating	The biggest steady current a fuse (or appliance) can safely carry — go above it and the fuse blows.	The maximum continuous current a component is designed to carry; for a fuse, the current at which it melts.	I ; amperes A	NOT the operating current itself (the rating is <i>above</i> it); NOT a voltage.	$I_{op} = 12.5 \text{ A} \Rightarrow$ choose 13 A (next standard rating up), not 3 A, not 30 A (Worked Ex 3).	Common trap: pick the fuse rating <i>just above</i> the operating current — too high a rating won't protect; too low blows in normal use.
Resistance from a power rating $R = V^2/P$	Working out a hidden resistance (like a kettle element) from its voltage and power label, without measuring current.	Rearrangement of $P = V^2/R$: $R = V^2/P$. Lets you find the element's resistance from its nameplate rating.	$R \text{ } \Omega$; $V \text{ V}$; $P \text{ W}$	NOT $R = P/V^2$; NOT a measured value — derived from the rating.	Kettle element: $R = V^2/P = 240^2/3000 = 19.2 \text{ } \Omega$ (fuse example, part b).	Common trap: $R = V^2/P$ (voltage <i>squared</i> on top); don't accidentally write $R = P/V^2$ or $R = V^2 P$.

2. Formula / case table

Quantity / case	Formula	Symbol meanings	Units	When to use	Rearrangements	Trap	Mini worked example
General power (master)	$P = VI$	P = power; V = p.d. across component; I = current through it	$P \text{ W}$; $V \text{ V}$; $I \text{ A}$	The default — whenever both V and I are given (or directly known). Works for <i>any</i> component.	$V = \frac{P}{I}$; $I = \frac{P}{V}$	Use the component's <i>own</i> V and I — for battery power use $V_{battery}$, not a resistor's p.d. (Phuc, a28d777a 55:43).	Lamp 6.0 V, 0.50 A $\Rightarrow P = 6.0 \times 0.50 = 3.0 \text{ W}$ (Self-test Q1a).
Power — heating effect	$P = I^2 R$	I = current; R = resistance	W	When I and R are given; for power <i>wasted as heat</i> in a resistor.	$I = \sqrt{P/R}$; $R = \frac{P}{I^2}$	$P \propto I^2$: doubling I quadruples P . Derive by $P = VI$ with $V = IR$.	0.75 A through 2.0 Ω : $P = (0.75)^2 \times 2.0 = 1.1 \text{ W}$ (Worked Ex 1).

Quantity / case	Formula	Symbol meanings	Units	When to use	Rearrangements	Trap	Mini worked example
Power — voltage form	$P = \frac{V^2}{R}$	V = p.d.; R = resistance	W	When V and R are given; avoids finding I first.	$V = \sqrt{PR}$; $R = \frac{V^2}{P}$	$P \propto V^2$ (fixed R): doubling V $\times 4$ power. $P \propto 1/R$ (fixed V): doubling R halves power.	150 Ω with 12 V: $P = 12^2/150 = 0.96$ W (Self-test Q3).
Formula-selection rule	given $I, V \Rightarrow P = VI$; $I, R \Rightarrow P = I^2 R$; $V, R \Rightarrow P = V^2/R$	choose by the two quantities you are handed	W	Every power calculation: underline the two given quantities, then pick.	(all three are the same physics via $V = IR$)	Don't substitute a value you <i>calculated</i> into a formula meant for <i>given</i> values without double-checking (Trap 4).	1.8 kW iron at 240 V: $I = P/V = 1800/240 = 7.5$ A (Self-test Q2).
Electrical energy (direct)	$W = Pt$	W = energy; P = power; t = time	W J; P W; t s	First choice for energy: when power and time are known. One step.	$P = \frac{W}{t}$; $t = \frac{W}{P}$	Use this <i>before</i> the $Q = It$, $E = QV$ roundabout — Phuc's documented pattern (Trap 1).	Lamp 3.0 W for 3.0 min: $W = 3.0 \times 180 = 540$ J (Self-test Q1b).
Electrical energy (IVt)	$W = VIt$	V = p.d.; I = current; t = time	J	Energy when V, I, t are the knowns (no separate power step).	$W = I^2 Rt = \frac{V^2}{R} t$	Equivalent to Pt ; pick the form matching your givens. Always quote the unit J.	Bulb 0.25 A, 2.2 V, 60 s: $W = 0.25 \times 2.2 \times 60 = 33$ J (Worked Ex 5).
Energy — heating ($I^2 Rt$)	$W = I^2 Rt$	I = current; R = resistance; t = time	J	Heat energy dissipated in a resistor over time t .	$W = Pt = VIt$	The energy version of the heating effect; same $\propto I^2$ scaling.	2.0 A in 5 Ω for 10 s: $W = 2^2 \times 5 \times 10 = 200$ J.

Quantity / case	Formula	Symbol meanings	Units	When to use	Rearrangements	Trap	Mini worked example
Charge-voltage energy	$E = QV$	Q = charge; V = p.d.	E J; Q C; V V	Only when the charge Q is explicitly given.	$Q = \frac{E}{V}$; $V = \frac{E}{Q}$	Phuc trap: $V = E/Q$ (energy per charge), NOT $V = E/t$ — time belongs in $I = Q/t$ (fb80c52f, Trap 3).	15 C through 2.2 V: $E = 15 \times 2.2 = 33$ J (roundabout route, avoid unless Q given).
Efficiency	$\eta = \frac{P_{useful}}{P_{total}} \times 100\%$	P_{useful} = useful output power; P_{total} = total input power (= VI electrical)	dimensionless → %	“How efficient is the device?”; comparing useful output to total input.	$\eta = \frac{W_{useful}}{W_{total}} \times 100\%$	Quote as a percentage ; label useful vs total; never $> 100\%$. $P_{total} = VI$ for electrical input.	In 50 W, useful 35 W $\Rightarrow \eta = 35/50 \times 100 = 70\%$ (Self-test Q7).
Fuse: operating current	$I = \frac{P}{V}$	P = appliance power rating; V = supply voltage	A	Step 1 of any fuse question: find the normal current.	$P = VI$	$I = P/V$, not V/P (Marta’s slip, e14ebf09).	Kettle 3.0 kW, 240 V: $I = 3000/240 = 12.5$ A (fuse example).
Fuse: choose rating	next standard rating above I (3 A / 5 A / 13 A)	standard UK ratings 3 A, 5 A, 13 A	A	Step 2: pick the fuse.	(compare I to the standard set)	Round up to the next rating; state the comparison (“12.5 < 13 A, so 13 A fuse”).	8.7 A hair-dryer $\Rightarrow$ 5 A would blow, fit 13 A (Self-test Q4).
Resistance from rating	$R = \frac{V^2}{P}$	V = rated voltage; P = rated power	Ω	Find a heating element’s resistance from its nameplate V and P .	$P = \frac{V^2}{R}$; $V = \sqrt{PR}$	V is squared on top; not P/V^2 .	Kettle: $R = 240^2/3000 = 57600/3000 = 19.2 \Omega$ (fuse example, part b).
Max safe current from power	$I = \sqrt{\frac{P_{max}}{R}}$	P_{max} = max safe power; R = resistance	A	Component rated for a maximum power; find the current limit.	from $P = I^2 R$	Expand prefixes first: 250 mW = 250×10^{-3} W (Trap 5).	250 mW, 47 k Ω : $I = \sqrt{0.25/47000} = 2.3$ mA (Worked Ex 4).

Quantity / case	Formula	Symbol meanings	Units	When to use	Rearrangements	Trap	Mini worked example
Definition of p.d.	$V = \frac{W}{Q}$	W = energy converted; Q = charge	$V = \text{J C}^{-1}$	Defining/explaining p.d.; energy per unit charge.	$W = QV$; $Q = W/V$	$V = W/Q$, NOT W/t (Phuc's recurring slip, Trap 3).	540 kJ, 2250 C $\Rightarrow V = 540000/2250 = 240 \text{ V}$ (kettle).
Ohm's law (for R)	$R = \frac{V}{I}$	V = p.d.; I = current	Ω	Finding resistance, or as the bridge that links the three power forms.	$V = IR$; $I = \frac{V}{R}$	$R = V/I$, NOT $R = IV$ ("Very Itchy Rash", Phuc/Marta slip e14ebf09 44:01, Trap 2).	12 V across element drawing 0.60 A: $R = 12/0.60 = 20 \Omega$.

3. Effect of scaling — one-glance table ($P \propto \dots$)

What you change	Formula in play	Effect on power P	Why
Double I (fixed R)	$P = I^2 R$	$P \times 4$	$P \propto I^2$ — squaring doubles twice over.
Double V (fixed R)	$P = V^2/R$	$P \times 4$	$P \propto V^2$.
Double R (fixed V)	$P = V^2/R$	$P \div 2$ (halved)	$P \propto 1/R$.
Double R (fixed I)	$P = I^2 R$	$P \times 2$ (doubled)	$P \propto R$.

4. Choosing the energy route — direct vs roundabout (Phuc's trap)

Route	Steps	Verdict
Direct	$W = Pt$ or $W = VIt$ — one substitution	Use this. Fewer steps, fewer unit-error chances; examiners reward the one-line method.
Roundabout	$Q = It$, then $E = QV$ — two substitutions	Avoid unless Q is explicitly given . Same answer, double the slip risk (Alex pushed the direct method, a28d777a 10:29).

5. Big picture (last-minute recall)

- Three power formulas are **one** equation in disguise: $P = VI$, then sub $V = IR$ to get $P = I^2R$, or $I = V/R$ to get $P = V^2/R$.
- **Pick by your givens:** $I, V \rightarrow VI$; $I, R \rightarrow I^2R$; $V, R \rightarrow V^2/R$.
- **Scaling:** double $I \Rightarrow \times 4$ power; double $V \Rightarrow \times 4$ power ($P \propto I^2$, $P \propto V^2$).
- **Energy:** go **direct** — $W = Pt = VIt = I^2Rt$. Only use $E = QV$ when charge Q is given.
- **Heating effect:** electrons collide with lattice ions $\Rightarrow$ ions vibrate more $\Rightarrow$ heat; rate $= I^2R$. Electrons are slowed, **not stored** (charge in = charge out).
- **Fuse:** $I = P/V$ (not V/P), then choose the **next standard rating above** (3 A / 5 A / 13 A).
- **Resistance from rating:** $R = V^2/P$ (voltage squared on top).
- **Efficiency:** $\eta = P_{\text{useful}}/P_{\text{total}} \times 100\%$ — always a percentage, never $> 100\%$.
- **Watch the slips:** $R = V/I$ not IV ; $V = W/Q$ not W/t ; expand prefixes (250 mW = 0.25 W) before substituting.